

The slide features a light gray background with abstract geometric shapes in blue and dark blue at the top and bottom. On the left side, there are several thin, black, circuit-like lines with small circles at their ends, extending horizontally and then turning at right angles. A similar set of lines is located on the right side.

Demand Estimation Lab

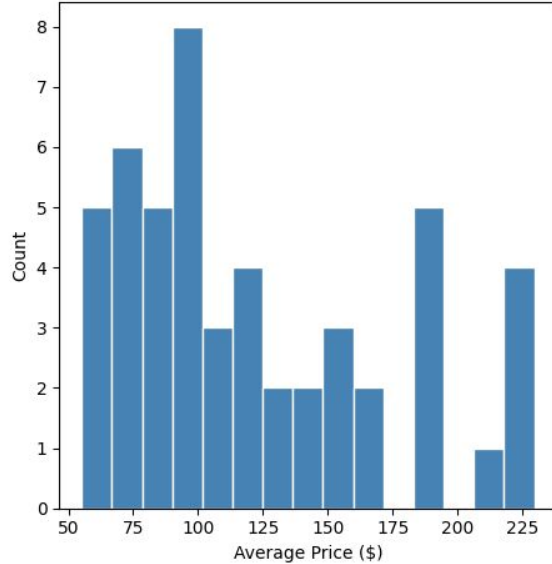
Adalia Kidanu, Betanya Binyam, Huda Mohmand

Market Overview

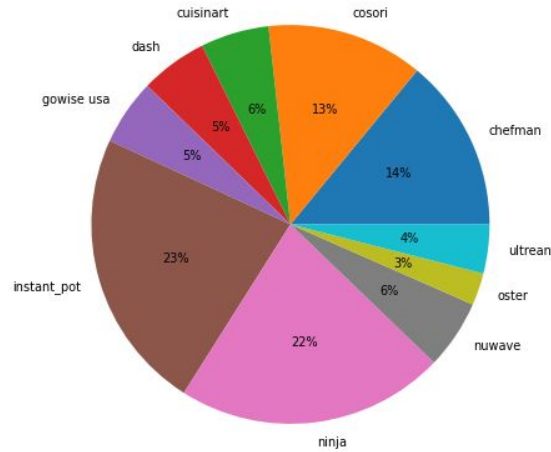
The air fryer market has grown rapidly over the past several years, driven by consumer demand for healthier, faster cooking options. On Amazon, 10 brands competed for market share between 2019 and 2023, with prices spanning from around \$55 for budget options like Dash to over \$229 for premium brands like Cuisinart. While Dash and Gowise USA captured large shares early on, Ninja and Instant Pot emerged as strong competitors over time. Products vary widely in form factor, with compact, oven style, and rotisserie models appealing to different consumer segments.

Market Trends

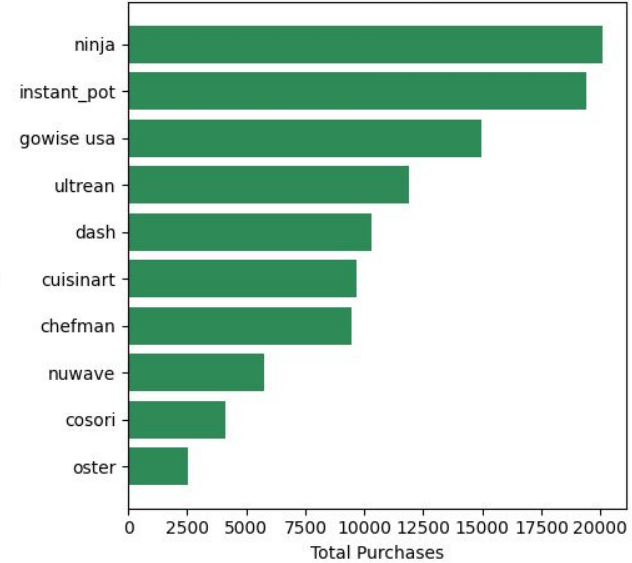
Price Distribution



Market Share (2023)



Total Purchases by Brand



Demand model

- We estimated air fryer demand using a linear regression demand model. The outcome variable was log brand market share, which measures each brand's relative demand in a given year. We used linear regression with average price, average rating, product feature shares, brand dummy variables, and year dummy variables as predictors. This helps estimate how demand changes with price and which product characteristics are associated with a stronger market performance.

Price Sensitivity

- Estimated price coefficient: -0.0377

The negative sign means that higher prices are associated with lower brand demand. When air fryer prices increase, brand market share tends to decrease. This matters because it confirms that the consumers of air fryers are price-sensitive and demand slope decreases. This negative price coefficient is also needed for pricing analysis because unit cost estimates depends on how demand responds to price.

Product Features & Differentiation

- Most common features:
 - Compact designs
 - Oven-style air fryers
- Rarer features:
 - Rotisserie features
 - Window features
 - Dual-basket designs
- Brand specialization:
 - Cuisinart, Dash, Oster, and Ultrean specialize heavily in oven-style models.
 - Chefman stands out for window and rotisserie features.
 - Ninja and Cosori are more compact-focused and less concentrated in oven-style or specialty features

The most common product feature in the air fryer market is compact design, which appears across almost every brand in this data set. Oven-style air fryers are also common, however, on the other hand dual-basket models are quite rare alongside window and rotisserie features. Brand specialization exists as well like how Chefman stands out for having some more window and rotisserie features. This showcases that the brands are not exactly competing with the exact same product designs, some brands position themselves around specific and more unique features.

Costs, Markups, and Profitability

- **Estimated unit costs**

Higher-cost brands: **Cuisinart, Oster, Ninja, Nuwave**

Lower-cost brands: **Dash, Ultrean, GoWise USA**

- **Markups (price – cost)**

Highest markups: **Ninja, Instant Pot, GoWise USA**

Most brands have **similar markups (~27–33)**

- **Share-weighted profits**

Most profitable brands: **Ninja, Instant Pot**

These brands combine **strong demand + solid markup**

Key Takeaways

- Profit depends on both **markup (margin)** and **market share (volume)**
- “High price alone doesn’t guarantee high profit”
- “Top brands succeed by balancing demand and pricing power”

Competitive Positioning & Conclusion

Overall Market Structure

- The market is **competitive but stable**
- Most brands stay in similar position over time

Brand Positioning

Ninja & Instant Pot → *Market leaders*

- High profits, strong demand, solid markups

GoWise USA → *High-volume competitor*

- Large market share, good overall profitability

Cuisinart & Oster → *Premium brands*

- Higher prices, but not the most profitable

Dash & Ultrean → *Budget brands*

- Lower prices, smaller margins, moderate share

Cosori & Nuwave → *Mid-tier / niche players*

- Moderate pricing, lower overall profits

Final takeaways

- Market leaders combine **strong demand and pricing power**
- Premium pricing does not always lead to higher profit
- Brands differentiate through price, features, and product focus

Overall, the brands that perform best are the ones that balance competitive pricing with strong demand, rather than relying only on premium pricing or extra features.